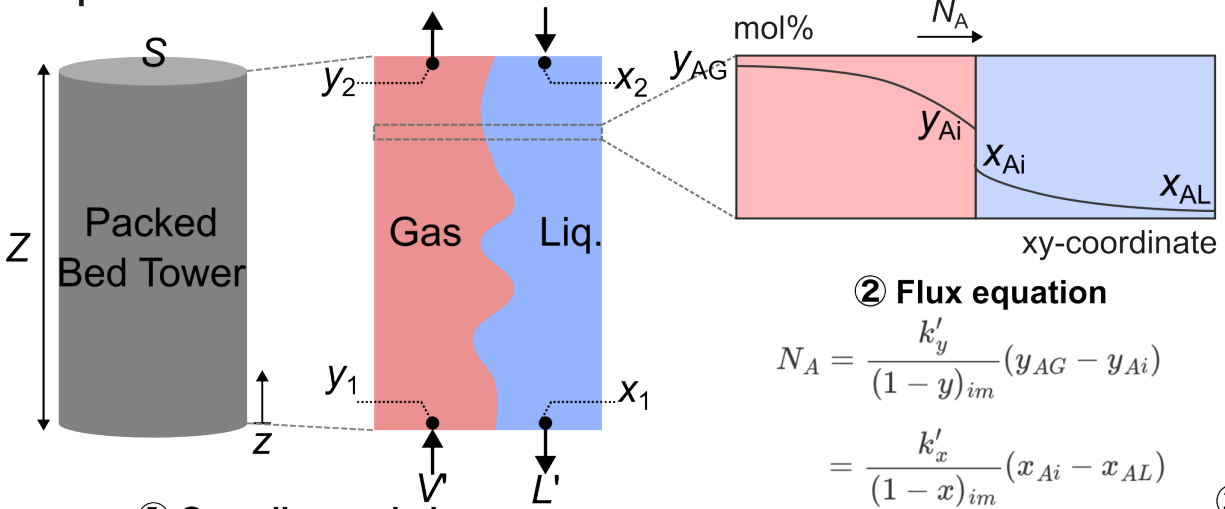
Cheatsheet: Absorption Tower

Absorption tower overview



① Overall mass balance

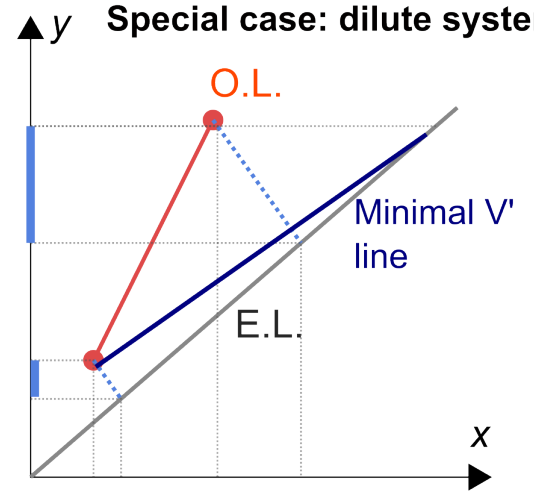
$$L'\left(\frac{x_2}{1-x_2}\right)+V'\left(\frac{y_1}{1-y_1}\right)=L'\left(\frac{x_1}{1-x_1}\right)+V'\left(\frac{y_2}{1-y_2}\right)$$

② Flux equation

$$N_A=\frac{k'_y}{(1-y)_{im}}(y_{AG}-y_{Ai})$$

$$=\frac{k'_x}{(1-x)_{im}}(x_{Ai}-x_{AL})$$

Special case: dilute system



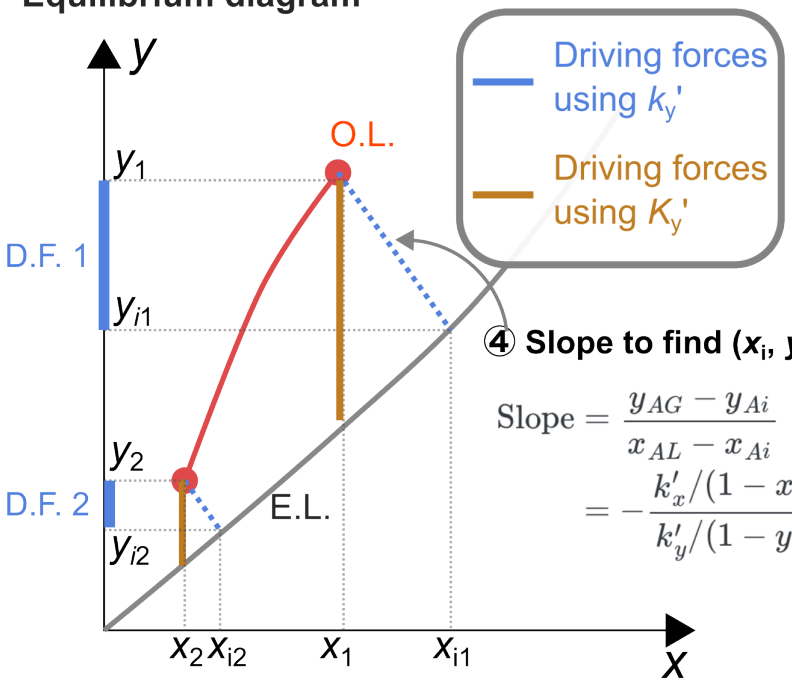
Conditions
- x < 0.1
- y < 0.1
- Linear line

Slope
L'/V'

③ Operating line slope (dilute)

$$y=\left(\frac{L'}{V'}\right)x+\left[y_1-\left(\frac{L'}{V'}\right)x_1\right]$$

Equilibrium diagram



④ Slope to find (x_i, y_i)

$$\text{Slope}=\frac{y_{AG}-y_{Ai}}{x_{AL}-x_{Ai}}$$

$$=-\frac{k'_x/(1-x)_{im}}{k'_y/(1-y)_{im}}$$

⑤ Height of tower

Gas

$$Z=\int_{y_2}^{y_1}\frac{V\,dy}{\frac{k'_y aS}{(1-y)_{im}}(1-y)(y-y_i)}$$

Liq.

$$Z=\int_{x_2}^{x_1}\frac{L\,dy}{\frac{k'_x aS}{(1-x)_{im}}(1-x)(x_i-x)}$$

Gas (dilute)

$$\frac{V}{S}(y_1-y_2)=k'_y aZ(y-y_i)_m$$

Liq. (dilute)

$$\frac{V}{S}(x_1-x_2)=k'_x aZ(x_i-x)_m$$

⑥ Use of log-mean forms

$$(1-y)_{im}=\frac{(1-y_{AG})-(1-y_{Ai})}{\ln\frac{1-y_{AG}}{1-y_{Ai}}}$$

$$(y-y_i)_m=\frac{(y_1-y_{i1})-(y_2-y_{i2})}{\ln\frac{y_1-y_{i1}}{y_2-y_{i2}}}$$